20 Solve algebraically

$$x^2 + y^2 = 18$$
$$x - 2y = -3$$

$x - 2y = -3$

$+2y \qquad +2y$

$x = 2y - 3$

Substitute: $(2y - 3)^2 + y^2 = 18$.

1 mark
Substitute for x

×	$2y$	-3
$2y$	$4y^2$	$-6y$
-3	$-6y$	$+9$

$4y^2 - 12y + 9$

1 mark
Expand correctly

$5y^2 - 12y + 9 = 18$

$-18 \qquad -18$

$5y^2 - 12y - 9 = 0$

1 mark
Quadratic = 0

×	y	-3
$5y$	$5y^2$	$-15y$
$+3$	$+3y$	-9

$(5y + 3)(y - 3) = 0$

1 mark
Factorise

$y = 3$ or $y = \dfrac{-3}{5} = -0.6$.

$x = 2(3) - 3 = 3$; $x = 2(-0.6) - 3 = -4.2$.

$(x, y) = (3, 3)$ or $(-4.2, -0.6)$.

1 mark
Both pairs

(Total for Question 20 is 5 marks)